

ZHIGALOVO, Russian Federation

WMO: 305210

Lat: 54.800N

Lon: 105.167E

Elev: 1375

StdP: 13.98

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.1	-40.1	-48.7	0.3	-43.7	-44.5	0.4	-39.7	16.7	4.3	12.6	3.0	0.1	300	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	86.6	64.7	82.9	63.3	79.2	62.0	68.1	79.3	66.3	76.9	64.6	74.9	4.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	96.0	71.6	62.8	89.8	70.3	60.8	83.6	68.4	33.3	79.4	31.8	76.6	30.5	75.1	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	13.5	10.6	DB	-47.0	93.8	5.5	3.2	-51.0	96.1	-54.2	98.0	-57.3	99.8	-61.3	102.1	(4)
(5)			WB	-46.3	70.9	5.0	2.3	-49.9	72.6	-52.8	74.0	-55.6	75.3	-59.3	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	-14.6	-5.6	14.6	34.3	46.4	60.0	65.7	60.4	46.3	29.7	6.2	-11.5
	DBStd	29.91	13.81	13.05	11.88	8.34	7.56	6.98	5.42	6.25	7.40	9.51	14.92	15.70
	HDD50	9314	2004	1557	1096	473	162	7	0	6	159	630	1314	1906
	HDD65	13698	2469	1977	1561	921	576	178	59	166	562	1094	1764	2371
	CDD50	1224	0	0	0	2	51	309	486	328	48	0	0	0
	CDD65	132	0	0	0	0	0	29	80	23	0	0	0	0
	CDH74	2165	0	0	0	0	85	658	988	413	21	0	0	0
	CDH80	660	0	0	0	0	14	202	332	109	3	0	0	0
Wind	WSAvg	3.2	1.7	2.2	3.4	4.8	5.2	3.8	3.2	3.2	3.4	3.3	2.6	1.8
Precipitation	PrecAvg	13.1	0.4	0.3	0.2	0.4	0.8	1.8	3.0	2.9	1.4	0.6	0.6	0.7
	PrecMax	18.8	1.2	0.9	0.6	1.1	1.7	3.9	7.6	5.1	2.9	1.5	1.1	1.3
	PrecMin	9.5	0.1	0.1	0.0	0.0	0.2	0.3	0.7	0.5	0.3	0.0	0.2	0.2
	PrecStd	2.1	0.2	0.2	0.2	0.2	0.4	0.8	1.6	1.1	0.6	0.4	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	30.5	48.0	67.3	82.4	91.0	91.8	88.2	76.6	58.8	37.7	25.6
		MCWB	21.8	26.5	37.1	49.0	58.5	64.1	66.9	65.7	59.0	46.7	33.9	23.9
	2%	DB	16.5	25.5	42.6	60.2	76.1	85.4	87.5	82.8	69.4	51.8	34.2	21.8
		MCWB	15.1	22.8	33.9	44.8	55.0	62.5	66.5	64.0	55.5	42.9	31.5	20.4
	5%	DB	10.7	19.7	37.5	54.6	70.5	81.5	83.7	78.7	63.9	46.5	30.6	17.3
		MCWB	9.6	17.8	30.9	41.7	52.3	61.4	65.0	62.3	52.9	41.2	28.3	16.1
	10%	DB	5.5	14.5	33.4	49.3	64.3	77.2	79.5	74.3	58.9	42.1	26.8	11.5
		MCWB	4.6	13.0	28.3	39.1	49.3	60.0	64.0	61.0	50.2	37.6	24.9	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	27.3	38.0	49.7	59.6	68.4	71.9	68.8	60.2	49.3	35.3	24.1
		MCDB	22.7	30.1	46.9	66.0	77.8	79.4	84.5	82.8	72.0	55.2	37.1	25.3
	2%	WB	15.1	22.9	34.3	45.8	56.2	66.0	69.2	65.9	56.8	44.5	31.4	20.3
		MCDB	16.4	25.7	41.7	57.4	72.6	78.1	81.1	77.7	66.8	49.8	33.6	21.6
	5%	WB	9.6	17.6	31.4	42.7	53.1	63.8	67.4	64.1	54.3	41.0	28.3	16.2
		MCDB	10.6	19.4	36.9	53.3	67.8	76.5	77.8	75.3	62.0	46.4	30.4	17.5
	10%	WB	4.7	12.9	28.6	39.5	50.4	61.6	65.8	62.1	51.5	38.0	24.8	10.6
		MCDB	5.5	14.5	33.2	48.4	63.0	74.2	75.6	71.7	57.4	41.7	26.6	11.5
Mean Daily Temperature Range		MDBR	17.1	25.3	29.6	23.6	26.7	26.3	23.6	22.8	21.2	17.7	16.8	15.8
		MCDBR	17.5	25.4	31.5	33.3	39.7	35.8	33.7	33.0	31.2	25.5	16.1	16.1
		MCWBR	16.7	22.2	23.1	18.7	19.8	14.6	13.8	14.6	16.2	16.5	13.7	15.0
	5% WB	MCDBR	17.3	25.0	29.3	30.7	35.7	28.4	25.6	27.6	28.1	23.5	15.6	16.4
		MCWBR	16.6	21.8	21.6	18.4	18.7	12.8	12.3	13.3	15.1	15.7	13.6	15.3
Clear-Sky Solar Irradiance	taub	0.276	0.305	0.329	0.365	0.406	0.441	0.454	0.394	0.341	0.322	0.294	0.272	
	taud	2.047	2.034	2.060	2.193	2.205	2.183	2.178	2.337	2.446	2.366	2.143	2.015	
	Ebn at Noon	199	238	263	271	266	256	250	260	262	237	192	168	
	Edh at Noon	22	32	39	40	42	43	43	35	27	23	20	18	
All-Sky Solar Radiation	RadAvg	314	639	1094	1508	1682	1828	1668	1424	960	555	297	213	
	RadStd	24	33	54	109	138	122	150	84	88	63	27	21	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		N/A	N/A	-3.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page